

The True Low Power MCU for Motor Control, Consumer and Industrial Applications



Renesas RL78/G14 Series 16 bit MCU is a high-performance rich function MCU of RL78 Family with the industry-leading low power consumption, realizing control of home-appliance motors and systems control by one chip. The G14 series features integrated motor control timers. Its on-chip oscillators with 1% accuracy provide an integrated low-cost solution for timing critical applications.



Key Features

- An ultra low power 16-bit MCU enabling lower system power consumption
- Enhanced real-time control capabilities
- 66 μ A/MHz @32MHz
- 0.57 μ A @ 32MHz with RTC and LVD
- New energy-saving SNOOZE mode: 0.7mA (UART), 1.2mA (ADC)
- Wide selection of memory capacities and packages suitable for consumer and industrial applications
- Memory sizes from 2KB to 512KB and packages sizes from 20 to 128 pins
- Conforms to European white goods safety standard (IEC 60730)
- Up to 1.39 DMIPS/MHz
- Flash memory with ECC (error-correcting code)

RL78/G14 Rich Functions

- **Timer RD (complimentary PWM mode)**
generates three-phase complementary PWM output ideal for driving brushless DC motors
- **Timer RG (phase counting mode)**
is ideal for counting in a two-phase encoder
- **Event link controller (ELC)**
activates other on-chip modules by using interrupt signals
- **Data transfer controller (DTC)**
- **High precision (+/-1%) internal oscillator**
over wide operation temperature